

Name: Answers

Exam Score: _____

1. (10pt) The following parts refer to $\mathbf{x}' = A\mathbf{x}$, where $A = \begin{bmatrix} 10 & -18 \\ 6 & -11 \end{bmatrix}$.

a) A change of variable $\mathbf{x} = P\mathbf{y}$ decouples the system $\mathbf{x}' = A\mathbf{x}$ and leads to the decoupled system $\mathbf{y}' = D\mathbf{y}$. Circle the correct pair for P and D .

(i) $P = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$

(ii) $P = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$

(iii) $P = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$

b) Circle the classification of the origin.

(i) attractor

(ii) repeller

(iii) saddle point

2. (16pt) For the following parts, let $\mathbf{w}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, $\mathbf{w}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{w}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{w}_4 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$

a) Circle the sets which are orthogonal.

(i) $\{\mathbf{w}_2, \mathbf{w}_3\}$

(ii) $\{\mathbf{w}_2, \mathbf{w}_4\}$

(iii) $\{\mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$

(iv) $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$

b) Circle the value of $\|\mathbf{w}_1\|$.

(i) 9

(ii) $\sqrt{3}$

(iii) 2

(iv) 3

(v) $\sqrt{2}$

c) Circle the projection of $\mathbf{w}_2$ onto the line containing $\mathbf{w}_1$.

(i) $\frac{5}{9} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

(ii) $\frac{5}{9} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

(iii) $\frac{5}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

(iv) $\frac{5}{3} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

3. (25pt) For the following parts, let $\mathbf{w}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{w}_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \\ 2 \end{bmatrix}$, $\mathbf{w}_3 = \begin{bmatrix} -2 \\ 4 \\ 0 \\ -2 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$.

a) Use the Gram-Schmidt algorithm to find an orthonormal basis for $W = \text{span}\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$

$$\begin{aligned} \mathbf{v}_1 &= \mathbf{w}_1, \quad \mathbf{v}_2 = \mathbf{w}_2 - \text{proj}_{\Omega_1}(\mathbf{w}_2), \quad \Omega_1 = \text{span}\{\mathbf{v}_1\} \\ &= \mathbf{w}_2 - \frac{\mathbf{v}_1 \cdot \mathbf{w}_2}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = \mathbf{w}_2 - \mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{v}_3 &= \mathbf{w}_3 - \text{proj}_{\Omega_2}(\mathbf{w}_3), \quad \Omega_2 = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\} \\ &= \mathbf{w}_3 - \frac{\mathbf{v}_1 \cdot \mathbf{w}_3}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\mathbf{v}_2 \cdot \mathbf{w}_3}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = \mathbf{w}_3 + 2\mathbf{v}_1 - \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \end{aligned}$$

$$\mathbf{u}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

b) Find $\text{proj}_{\Omega}(\mathbf{b})$ and $\text{proj}_{\Omega^\perp}(\mathbf{b})$ where $\Omega = \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$.

$$\Omega = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\} \Rightarrow \text{proj}_{\Omega}(\mathbf{b}) = \frac{\mathbf{v}_1 \cdot \mathbf{b}}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\mathbf{v}_2 \cdot \mathbf{b}}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 = \frac{1}{2} \mathbf{v}_1 + \frac{1}{2} \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{proj}_{\Omega^\perp}(\mathbf{b}) = \mathbf{b} - \text{proj}_{\Omega}(\mathbf{b}) = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Alternatively, with $A = [\mathbf{w}_1, \mathbf{w}_2]$

$$\text{proj}_{\Omega}(\mathbf{b}) = A(A^T A)^{-1} A^T \mathbf{b} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

4. (24pt) For the following parts, let

$$A = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}, \quad A^T A = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 3 & -3 \\ 2 & -3 & 5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

a) Find the least squares solutions for $A\mathbf{x} = \mathbf{b}$.

$$\left[A^T A \mid A^T \mathbf{b} \right] = \left[\begin{array}{ccc|c} 2 & 0 & 2 & 1 \\ 0 & 3 & -3 & 0 \\ 2 & -5 & 5 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1/2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \begin{array}{l} x_1 + x_3 = 1/2 \\ x_2 - x_3 = 0 \end{array}$$

$$\Rightarrow \mathbf{x} = \begin{bmatrix} 1/2 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \quad x_3 \in \mathbb{R}$$

the l.s.s

b) Find the vector $\text{proj}_{\text{Col}(A)}(\mathbf{b})$.

$$A \begin{bmatrix} 1/2 \\ 0 \\ 0 \end{bmatrix} = 1/2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

c) Find the distance between $(0, 1, -1)$ and $\text{Col}(A)$.

$$\mathbf{b} - \text{proj}_{\text{Col}(A)}(\mathbf{b}) = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} - 1/2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \\ -1 \end{bmatrix} = 1/2 \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$$

$$\|\mathbf{b} - \text{proj}_{\text{Col}(A)}(\mathbf{b})\| = 1/2 \sqrt{6} = \underline{\text{distance}}$$

5. (11pt) For the following parts, $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$.

a) Circle the number for the matrix pair that gives an orthogonal diagonalization of A .

(i) $P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

(ii) $P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

☒ (iii) $P = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

(iv) $P = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, D = \begin{bmatrix} -3 & 0 \\ 0 & 1 \end{bmatrix}$

b) Circle all equations that relate A , P and D in part a).

☒ (i) $PD = AP$ $\leftarrow A = PDP^T$ however $\underline{P^{-1} = P^T}$

(ii) $A = P^T DP$

(iii) $D = PAP^T$

☒ (iv) $A = PDP^T$

6. (14pt) For the following parts $Q(x_1, x_2) = 3x_1^2 + 8x_1x_2 + 3x_2^2 = \mathbf{x}^T A \mathbf{x}$. A is the matrix for the quadratic form Q .

a) Circle the matrix A for the quadratic form Q .

(i) $A = \begin{bmatrix} 3 & 8 \\ 8 & 3 \end{bmatrix}$

☒ (ii) $A = \begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix}$

(iii) $A = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}$

(iv) $A = \begin{bmatrix} 8 & 3 \\ 3 & 8 \end{bmatrix}$

$\Rightarrow (\lambda - 3)^2 - 16$ (char. poly)
 $\Rightarrow$ e. values are 7, -1

b) Classify the quadratic form Q by circling one of the following:

(i) positive definite

(ii) positive semidefinite

☒ (iii) indefinite

(iv) negative semidefinite

(v) negative definite

c) Provide values for the following:

(i) $\min_{\|\mathbf{x}\|=1} \mathbf{x}^T A \mathbf{x} = \underline{-1}$

(ii) $\max_{\|\mathbf{x}\|=1} \mathbf{x}^T A \mathbf{x} = \underline{7}$